Heterogeneous Verification for Multi-Step Decentralized Workflows

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Abstract

Verifiable multi-step automation would extend blockchain guarantees beyond single-transaction execution, enabling multi-step workflows with cryptographic attestations. Smart contracts provide verification for onchain computation, but these guarantees are lost when workflows require offchain execution such as external APIs, confidential compute, or nondeterministic operations. We propose a heterogeneous verification system in which each workflow step uses the verification mechanism appropriate to its computation. The system composes cryptographic proofs for blockchain operations, hardware attestations for confidential execution, and stake-backed attestations for external interactions into unified workflow attestations. Each step produces a signed attestation that validators broadcast and stake against. Provided honest validators control sufficient stake, fraudulent attestations can be challenged via onchain fraud proofs, preserving end-to-end verifiability across heterogeneous trust assumptions. The system enables verifiable automation previously impossible under uniform trust models.

1 Introduction

Blockchain systems establish cryptographic verification for single transaction operations, enabling trustless execution of financial transfers and smart contracts. While this works well for onchain computation, verification guarantees are lost when automation requires offchain execution. Multi-step workflows that interact with external APIs, process confidential compute, or perform nondeterministic operations cannot maintain end-to-end verifiability under current blockchain architectures. Developers face a binary choice: constrain all computation to onchain within gas limits and block times, or move computation offchain and abandon cryptographic verification. This forces workflows to trade functionality for verification or verification for functionality. Existing oracle networks provide partial, domain-specific feeds, but no general mechanism exists for verifiable multi-step automation across heterogeneous trust environments.

What is needed is a verification system that adapts to each computation's trust properties rather than imposing uniform assumptions. Cryptographic proofs verify blockchain operations, hardware attestations protect confidential execution, and stake-backed attestations cover external interactions, composing into unified workflow attestations. We propose a heterogeneous verification system in which validators attest to each workflow step using the appropriate mechanism and stake against their attestations; fraudulent attestations can be challenged via onchain fraud proofs. The system maintains verifiability provided honest validators control sufficient stake to make fraud economically irrational.

References

[1] S. Nakamoto, "Bitcoin: A Peer-to-Peer Electronic Cash System," 2008.